

Q.1 Let $X_1 \sim N(1, 1)$

$$X_2 | X_1 \sim N(1 + \rho(X_1 - 1), 1 - \rho^2)$$

$$f_{X_1, X_2}(x_1, x_2) = f_{X_1}(x_1) f_{X_2 | X_1}(x_2 | x_1)$$

$$= \frac{1}{\sqrt{2\pi}} \exp\left\{-\frac{1}{2}(x_1 - 1)^2\right\} \cdot \frac{1}{\sqrt{2\pi(1-\rho^2)}} \exp\left\{-\frac{1}{2(1-\rho^2)}(x_2 - 1 - \rho(x_1 - 1))^2\right\}$$

$$= \frac{1}{2\pi\sqrt{1-\rho^2}} \exp\left[-\frac{1}{2} \left\{ \frac{(1-\rho^2)(x_1 - 1)^2 + (x_2 - 1)^2 - 2\rho(x_1 - 1)(x_2 - 1) + \rho^2(x_1 - 1)^2}{1-\rho^2} \right\}\right]$$

$$= \frac{1}{2\pi\sqrt{1-\rho^2}} \exp\left[-\frac{1}{2} \left\{ \frac{(x_1 - 1)^2 + (x_2 - 1)^2 - 2\rho(x_2 - 1)(x_1 - 1)}{1-\rho^2} \right\}\right]$$

$$f_{X_2}(x_2) = \int f_{X_1, X_2}(x_1, x_2) dx_1$$

$$= \frac{1}{2\pi\sqrt{1-\rho^2}} \int \exp\left[-\frac{1}{2} \left\{ \frac{(x_1 - 1)^2 + (x_2 - 1)^2 - 2\rho(x_2 - 1)(x_1 - 1)}{1-\rho^2} \right\}\right] dx_1$$

$$= \frac{\exp\left\{-\frac{1}{2}(x_2 - 1)^2\right\}}{\sqrt{2\pi}} \times$$

$$\int_{-\infty}^{\infty} \frac{1}{\sqrt{2\pi(1-\rho^2)}} \exp\left\{-\frac{1}{2(1-\rho^2)} [x_1 - (1 + \rho(x_2 - 1))]^2\right\} dx_1$$

$$= \frac{\exp\left\{-\frac{1}{2}(x_2 - 1)^2\right\}}{\sqrt{2\pi}}, \quad x_2 \in \mathbb{R}.$$

$\Uparrow$
 $= 1$

$$\therefore x_2 \sim N(1, 1)$$

(2) Exponential distribution as a prior of θ is not possible. $\left[\frac{1}{2} \text{ marks}\right]$

$$\text{If } \theta \sim \text{Exp} \left(\text{mean} = \frac{1}{\lambda} \right)$$

$$\text{i.e., } \pi(\theta) = \begin{cases} \lambda e^{-\lambda\theta} & , \theta > 0 \\ 0 & \text{o.w.} \end{cases}$$

$$\text{Then } \mathbb{E}[\theta] = \frac{1}{\lambda} \quad , \quad \text{Var}[\theta] = \frac{1}{\lambda^2} = 3 \quad (\text{as given})$$

$$\text{It's not possible to have, } \frac{1}{\lambda} = 2 \Rightarrow \frac{1}{\lambda^2} = 4 \neq 3$$

• If one can choose $\theta \sim G(a, b)$

$$\text{Then } \mathbb{E}[\theta] = \frac{a}{b} \quad , \quad \text{Var}[\theta] = \frac{a}{b^2}$$

$$\Rightarrow \frac{a}{b} = 2 \quad , \quad \frac{a}{b^2} = 3$$

$$\Rightarrow a = \frac{4}{3} \quad , \quad b = \frac{2}{3}$$

$\left[2 \text{ marks for correct parameters with distributions} \right]$

• If one choose $\theta \sim \text{IG}(a, b)$. with $a > 2$

$$\therefore \mathbb{E}[\theta] = \frac{b}{a-1}, \quad \text{Var}[\theta] = \frac{b^2}{(a-1)^2(a-2)}$$

$$\Rightarrow a = \frac{10}{3}, \quad b = \frac{14}{3}$$

[1 mark deduction for wrong priors
1 mark deduction for wrong parameters]

[3]

Let θ_1 and θ_2 denote the click rate per hour for the old and new websites:

If Y_1 and Y_2 denote the total no. of observed clicks respectively, the likelihood is given by:

$$Y_i | \theta_i \stackrel{\text{ind}}{\sim} \text{Poi}(30 \times 24 \theta_i = 720 \theta_i), i=1, 2.$$

The likelihood is given by,

$$\begin{aligned} L(\theta_1, \theta_2 | Y_1, Y_2) &= \prod_{i=1}^2 \frac{1}{Y_i!} (720 \theta_i)^{Y_i} e^{-720 \theta_i} \\ &\propto \theta_1^{Y_1} \theta_2^{Y_2} \exp\{-720(\theta_1 + \theta_2)\} \end{aligned}$$

[Wrong likelihood: -1 marks]

Conjugate priors: $\theta_i \stackrel{\text{ind}}{\sim} G(a_i, b_i), a_i, b_i > 0$
 $i=1, 2$

(a_1, b_1, a_2, b_2) are known.

Posterior of (θ_1, θ_2)

$$\therefore \pi(\theta_1, \theta_2 | Y_1, Y_2) \propto L(\theta_1, \theta_2 | Y_1, Y_2) \pi(\theta_1, \theta_2)$$

$$\propto \theta_1^{y_1} \theta_2^{y_2} \exp \left\{ -720(\theta_1 + \theta_2) \right\}$$

$$\times \frac{\theta_1^{a_1-1} b_1^{a_1} e^{-b_1 \theta_1}}{\Gamma(a_1)} \times \frac{\theta_2^{a_2-1} b_2^{a_2} e^{-b_2 \theta_2}}{\Gamma(a_2)}$$

$$\propto \theta_1^{a_1+y_1-1} \theta_2^{a_2+y_2-1} e^{-(720+b_1)\theta_1} e^{-(720+b_2)\theta_2}$$

$$\therefore \theta_1 | Y_1 \sim G(a_1 + Y_1, b_1 + 720)$$

$$\theta_2 | Y_2 \sim G(a_2 + Y_2, b_2 + 720)$$

0.5

Now, to check whether the new websites provide a better reach, one need to calculate:

$$\pi(\theta_1 < \theta_2 | Y_1, Y_2)$$

$$= \int \int_{0 < \theta_1 < \theta_2 < \infty} \pi(\theta_1, \theta_2 | Y) d\theta_1 d\theta_2 = p \text{ (say)}$$

- No analytical expression is possible.

Algorithm:

1

- Draw n iid samples: (n is large)

$$\tilde{\theta}^{(1)} = (\theta_{11}, \theta_{12}, \dots, \theta_{1n}) \text{ from } G_1(a_1 + \gamma_1, b_1 + \gamma_2)$$

$$\tilde{\theta}^{(2)} = (\theta_{21}, \theta_{22}, \dots, \theta_{2n}) \text{ from } G_1(a_2 + \gamma_2, b_2 + \gamma_2)$$

- Define, $z_i = \theta_{1i} - \theta_{2i}$

$$\text{Define; } m_i = \begin{cases} 1 & ; z_i < 0 \\ 0 & , z_i > 0 \end{cases}$$

$$\text{Calculate: } \bar{m} = \frac{1}{n} \sum_{i=1}^n m_i \xrightarrow{\text{a.s.}} p$$

- [Partial wrong likelihood with not γ_2 will award with wrong posterior - 1 marks]

- No partial awarding for wrong algorithm.

4. θ : the success probability of shooting a balloon at a fair at Kanpur.

Y : The no. of trials of getting the first success.

So the correct probability model is:

$$Y|\theta \sim \text{Geo}(\theta), \quad 0 < \theta < 1$$

$$\therefore f(y) = \begin{cases} (1-\theta)^{y-1} \theta & , y = 1, 2, 3, \dots \\ 0 & \text{o.w.} \end{cases}$$

$$[\text{Not } (1-\theta)^y \theta, y = 0, 1, 2, \dots] \times$$

[Correct model : 1 marks]

$$\theta \sim \text{Beta}(m, n), \quad m, n > 0 \text{ (known)}.$$

• Posterior: $\theta|Y \sim \text{Beta}(m+1, n+y-1)$

$$\therefore \mathbb{E}[\theta|Y=y] = \frac{m+1}{m+n+y}$$

$$\text{Var}[\theta|Y=y] = \frac{(m+1)(n+y-1)}{(m+n+y)^2(m+n+y+1)}$$

[Correct posterior mean: 1 marks]

Given $Y=y$, the 95% credible interval for θ , say $[\xi_{0.025}, \xi_{0.975}]$ is given as:

where,

$$\left. \begin{aligned} \xi_{0.025} &= \text{pbeta}(0.025, \text{shape1} = m+1, \text{shape2} = n+y-1) \\ \xi_{0.975} &= \text{pbeta}(0.975, \text{shape1} = m+1, \text{shape2} = n+y-1) \end{aligned} \right\} \text{1 mark}$$

Or, estimate it by n posterior samples.

$$\mathbb{P}[\xi_{0.025} < \theta < \xi_{0.975} \mid Y] = 0.95$$

$$\Rightarrow \mathbb{P}[\theta < \xi_{0.975} \mid Y] - \mathbb{P}[\theta < \xi_{0.025} \mid Y] = 0.95.$$

Now, for any p , ξ_p

$$\mathbb{P}[\theta < \xi_p \mid Y] = \int_0^{\xi_p} \frac{1}{\beta(m+1, n+y-1)} \theta^m (1-\theta)^{n+y-2} d\theta$$

$$\xi_p = F^{-1}(p)$$

Empirical cdf: $\hat{F}(t) = \frac{1}{n} \sum_{i=1}^n I(\theta_i \leq t)$

and $\hat{\xi}_p = \hat{F}^{-1}(p)$

[Exact expression with ECDF will give
1 marks]

- Partial (0.5) marks for mentioning the quantiles.